STAT6611

- Lecture 18 (T, Oct 27)
- solutions posted to Homework 3
- Midterm quiz (last 15-10 min)
- No new homework
  (next assignment R, Oct 29)
  Homework 4

Lemma 3.2. *An $n \times n$ symmetric matrix A is idempotent of rank r if and only if it has the characteristic root 1 with multiplicity r and the characteristic root 0 with multiplicity $n - r$.*

In other words, A symmetric/idempotent ($A^2 = A$) $\Leftrightarrow A = P \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix} P'$ orthogonal P ($P'P = I_n$)

Proof. Corresponding to an $n \times n$ symmetric matrix A of rank r, there exists an orthogonal matrix P such that

$$P'AP = D,$$

i.e. $A = P \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix} P'$ holds for D_r with non-zero entries

where D is a diagonal matrix whose diagonal elements $d_1, \ldots, d_n$ are the characteristic roots of A. Necessarily, $\text{rank}(D) = r$, that is, D has r nonzero elements.

A is idempotent $\Leftrightarrow A^2 = A$

$\Leftrightarrow P \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix} P'P \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix} P' = P \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix} P'$ ($P'P = I$)

$\Leftrightarrow \begin{bmatrix} D_r^2 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix} \Leftrightarrow D_r^2 = D_r$ (D_r^{-1} exists)

$\Leftrightarrow D_r = I_r$

Lemma 3.3. *Take $r_1(x), r_2(x), s_1(x)$, and $s_2(x)$ to be polynomials in a real variable x. Suppose that*

$$r_2(x) \equiv \gamma(x - \lambda_1)^{m_1}(x - \lambda_2)^{m_2} \cdots (x - \lambda_k)^{m_k},$$

where k is a nonnegative integer, $m_1, \ldots, m_k$ are positive integers, $\lambda_1, \ldots, \lambda_k$ are constants, and γ is a nonzero constant. Suppose further that

$$\frac{s_1(x)}{s_2(x)} = \exp\left[\frac{r_1(x)}{r_2(x)}\right] \quad \text{for} \quad x \in J,$$

where J is some nondegenerate interval that does not include $\lambda_1, \ldots, \lambda_k$ or any roots of $s_2(x)$. Then, there exist constants α and β such that $r_1(x) = \alpha r_2(x)$ and $s_1(x) = \beta s_2(x)$ for all x.

The proof of Lemma 3.3 is discussed in the paper by Driscoll and Gundberg, "A history of the development of Craig's theorem" (*The American Statistician*, Volume 40, 1986, 65–70).

ie Σ^{-1} exists

Theorem 3.16. *Let* $\underline{x} \sim N(\underline{\mu}, \Sigma)$, *where* $\underline{x}$ *is an* $r \times 1$ *vector and* $\text{rank}(\Sigma) = r$. *Let* A *be an* $r \times r$ *symmetric matrix of constants. If* $A\Sigma$ *is idempotent of rank* $s \geq 1$, *then* $\underline{x}'A\underline{x} \sim \chi^2\left(s, \frac{1}{2}\underline{\mu}'A\underline{\mu}\right)$. *Conversely, if* $\underline{x}'A\underline{x} \sim \chi^2(s, \lambda)$ *for some* λ, *then* $A\Sigma$ *is idempotent of rank* s.

this characterizes what is needed for a quadratic form $\underline{x}'A\underline{x}$ to chi-square distributed (possibly non-central)
We'll later see that "Σ" must be a generalized inverse of A (i.e. $A\Sigma A = A$). Also, note $s = \text{rank}(A\Sigma) = \text{rank}(A) \leq r$.

We first prove the theorem for the case $\Sigma = I$, which simplifies as follows.

Let $\underline{x} \sim N(\underline{\mu}, I_r)$. *Let* A *be an* $r \times r$ *symmetric matrix of constants. If* A *is idempotent of rank* $s \geq 1$, *then* $\underline{x}'A\underline{x} \sim \chi^2\left(s, \frac{1}{2}\underline{\mu}'A\underline{\mu}\right)$. *Conversely, if* $\underline{x}'A\underline{x} \sim \chi^2(s, \lambda)$ *for some* λ, *then* A *is idempotent of rank* s.

Proof. Suppose first $r \times r$ A is idempotent of rank r.

$\Rightarrow$ If $r \times r$ A has $\text{rank}(A) = r$ & A is idempotent ($A^2 = A$)
$\Rightarrow$ since A^{-1} invertible, $A = I_r$
Then, $\underline{x}'A\underline{x} = \underline{x}'\underline{x} \sim \chi^2(r, \frac{1}{2}\underline{\mu}'\underline{\mu})$ ($\underline{x} \sim N(\underline{\mu}, I_r)$); $\frac{1}{2}\underline{\mu}'A\underline{\mu}$

If A is idempotent of rank s where $1 \leq s < r$, then by Lemma 3.2, there exists an orthogonal matrix $P = (P_1, P_2)$, where P_1 is an $r \times s$ matrix, such that

$$A = P \begin{bmatrix} I_s & 0 \\ 0 & 0 \end{bmatrix} P' = (P_1, P_2) \begin{bmatrix} I_s & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} P_1' \\ P_2' \end{bmatrix} = (P_1, 0) \begin{bmatrix} P_1' \\ P_2' \end{bmatrix} = P_1 P_1'.$$

$r \times r$ (P); $r \times s$; $s \times r$; $r \times s$ (P_1, P_2); $r \times s$ P_1; $s \times r$ P_1'; $r \times s$ $s \times r$

Lemma 3.2

$P'P = I_r$ $P_{r \times r}$

Note that

$$\begin{bmatrix} I_s & 0 \\ 0 & I_{r-s} \end{bmatrix} = I_r = P'P = \begin{bmatrix} P_1' \\ P_2' \end{bmatrix} (P_1, P_2) = \begin{bmatrix} P_1'P_1 & P_1'P_2 \\ P_2'P_1 & P_2'P_2 \end{bmatrix},$$

which implies that $P_1'P_1 = I_s$. Let $\underline{z} = P_1'\underline{x}$. Then, $\underline{x}'A\underline{x} = \underline{x}'P_1P_1'\underline{x} = \underline{z}'\underline{z}$.

Next suppose (still assuming $\underline{x} \sim N(\underline{\mu}, I_r)$) that $\underline{x}'A\underline{x} \sim \chi^2(s, \lambda)$ for some λ. Need to show A is idempotent of rank s.

Let $m = \text{rank}(A)$. Since A is symmetric, for an orthogonal matrix $P = (P_1, P_2)$, where P_1 is an $r \times m$ matrix, we can write

$$A = P \begin{bmatrix} D_m & 0 \\ 0 & 0 \end{bmatrix} P' = (P_1, P_2) \begin{bmatrix} D_m & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} P_1' \\ P_2' \end{bmatrix} = P_1 D_m P_1'.$$

where D_m is a diagonal matrix with entries (say) $d_1, \ldots, d_m$.

Since $P_1'P_1 = I_m$ and since $\underline{x} \sim N(\underline{\mu}, I_r)$, we have $\underline{z} = P_1'\underline{x} \sim N(P_1'\underline{\mu}, I_m)$. Note

$$\underline{x}'A\underline{x} = \underline{x}'P_1 D_m P_1'\underline{x} = \underline{z}'D_m\underline{z} = \sum_{i=1}^{m} d_i z_i^2, \quad \underline{z} = \begin{pmatrix} z_1 \\ \vdots \\ z_m \end{pmatrix}.$$

$$M_{\underline{x}'A\underline{x}}(t) = \mathrm{E}\exp\left(t\underline{x}'A\underline{x}\right) = \mathrm{E}\exp\left(t\sum_{i=1}^{m} d_i z_i^2\right) = \mathrm{E}\prod_{i=1}^{m}\exp\left(td_i z_i^2\right)$$
$$= \prod_{i=1}^{m}\mathrm{E}\exp\left(td_i z_i^2\right) = \prod_{i=1}^{m} M_{z_i^2}(td_i)$$
$$= \prod_{i=1}^{m}\left\{(1-2d_i t)^{-1/2}\exp\left(\frac{2td_i\frac{1}{2}\mu_{z_i}^2}{1-2d_i t}\right)\right\}$$

provided t satisfies $td_i < 1/2$ for each $i = 1, \ldots, m$.

By assumption, though, $\underline{x}'A\underline{x} \sim \chi^2(s, \lambda)$ for some λ, so it must be

$$M_{\underline{x}'A\underline{x}}(t) = \left\{(1-2t)^{-s/2}\exp\left(\frac{2t\lambda}{1-2t}\right)\right\}, \quad t < 1/2.$$

So for small t around 0 (where $|t| < 1/2$ and $|t|\min_{1\le i\le m}|d_i| < 1/2$), we have

$$\left\{(1-2t)^{-s/2}\exp\left(\frac{2t\lambda}{1-2t}\right)\right\} = \prod_{i=1}^{m}\left\{(1-2d_i t)^{-1/2}\exp\left(\frac{2td_i\frac{1}{2}\mu_{z_i}^2}{1-2d_i t}\right)\right\},$$

which can be re-arranged algebraically as

$$\frac{\prod_{i=1}^{m}(1-2d_i t)}{(1-2t)^s} = \exp\left(\frac{-4t\lambda}{1-2t} + 4t\sum_{i=1}^{m}\frac{d_i\frac{1}{2}\mu_{z_i}^2}{1-2d_i t}\right).$$

By Lemma 3.1, a constant β exists where $\prod_{i=1}^{m}(1-2d_it) = \beta(1-2t)^s$ for all $t \in \mathbb{R}$.

For $t=0$, $1 = \beta \cdot 1$ or $\beta = 1$

It must be that $m = s$ (degrees match)

& roots of RHS (roots at $t = \frac{1}{2}$ always)

= roots of LHS (roots at $t = \frac{1}{2d_i}$)

$\Rightarrow d_i = 1, i = 1, \ldots, s = m$

We now prove the theorem for the case Σ is nonsingular (by assumption) but $\Sigma \neq I$.

$\uparrow$ $Var(\underline{x}) = \Sigma$ $r \times r$ matrix with rank r

Express $\Sigma = \Gamma'\Gamma$, where Γ is a nonsingular matrix, and let $\underline{w} = \left(\Gamma'\right)^{-1}\underline{x}$. Then,

$\uparrow$ Γ has rank $(\Sigma) = r$ (so Γ is invertible)

$$\underline{x}'A\underline{x} = \underline{w}'\Gamma A\Gamma'\underline{w} \quad \text{and} \quad \underline{w} \sim N\left[\left(\Gamma'\right)^{-1}\underline{\mu}, I\right].$$

quad form in $\underline{w}$ $\uparrow$ $Var(\underline{w}) = (\Gamma')^{-1} Var(\underline{x}) (\Gamma^{-1}) = I_r$

Thus, we can apply the theorem for the case $\Sigma = I$ already proved above to $\underline{w}'\left(\Gamma A\Gamma'\right)\underline{w}$. That is, if $\Gamma A\Gamma'$ is idempotent of rank $s \geq 1$ (note that $\Gamma A\Gamma'$ is symmetric), then

$$\underline{w}'\left(\Gamma A\Gamma'\right)\underline{w} \sim \chi^2\left[s, \frac{1}{2}\underline{\mu}'\Gamma^{-1}\left(\Gamma A\Gamma'\right)\left(\Gamma'\right)^{-1}\underline{\mu}\right], \text{ that is, } \underline{x}'A\underline{x} \sim \chi^2\left(s, \frac{1}{2}\underline{\mu}'A\underline{\mu}\right);$$

and, conversely, if $\underline{x}'A\underline{x} \sim \chi^2(s,\lambda)$ for some λ, that is, if $\underline{w}'\left(\Gamma A\Gamma'\right)\underline{w} \sim \chi^2(s,\lambda)$ for some λ, then $\Gamma A\Gamma'$ is idempotent of rank s.

quad form

It remains to show that $\Gamma A\Gamma'$ is idempotent of rank s if and only if $A\Sigma$ is idempotent of rank s.

$r \times r$ Γ is invertible, so $\text{rank}(\Gamma A\Gamma') = \text{rank}(A\Gamma'\Gamma)$

Here, $\text{rank}(\Gamma A\Gamma') = s$ Γ^{-1} exist $= \text{rank}(A\Sigma)$

$\Leftrightarrow \text{rank}(A\Sigma) = s$.

$\Gamma A\Gamma'$ idempotent $\Leftrightarrow \Gamma A\Gamma'\Gamma A\Gamma' = \Gamma A\Gamma'$ $\Leftrightarrow$ (Γ^{-1}) $A\Gamma'\Gamma A\Gamma'\Gamma = A\Gamma'\Gamma$

$\Leftrightarrow A\Sigma A\Sigma = A\Sigma$

$\Leftrightarrow A\Sigma$ is idempotent

Remark 1. Assuming Σ is nonsingular, we have

$$A\Sigma = A\Sigma A\Sigma \iff A = A\Sigma A \iff \Sigma A = \Sigma A\Sigma A,$$

that is, $A\Sigma$ is idempotent $\iff$ Σ is a generalized inverse of A $\iff$ ΣA is idempotent.

Remark 2. A generalization of Theorem 3.16 that covers the second-degree polynomial statistic $\underline{x}'A\underline{x}+\underline{a}'\underline{x}+a_0$, as well as the quadratic form $\underline{x}'A\underline{x}$, and that covers singular Σ, as well as nonsingular Σ, can be found, for example, in Searle's book *Linear Models* (see his Theorem 2s).

Example 1. Consider Model II and suppose that $\underline{y}$ has a multivariate normal distribution. Then, $\underline{y} \sim N(X\underline{\beta}, \sigma^2 I)$. Let $\widehat{\underline{\beta}}$ be any solution to the normal equations $X'X\widehat{\underline{\beta}} = X'\underline{y}$, and let $P_X = X(X'X)^- X'$. It can be shown that

$$\widehat{\sigma}^2 = (\underline{y} - X\widehat{\underline{\beta}})'(\underline{y} - X\widehat{\underline{\beta}})/(n-p^*) = \underline{y}'(I-P_X)\underline{y}/(n-p^*)$$

is the uniformly minimum variance unbiased estimator (UMVUE) of σ^2.

Thus, by Theorem 3.16,

$$\left(\frac{n-p^*}{\sigma^2}\right)\widehat{\sigma}^2 = \underline{y}'\left[(1/\sigma^2)(I-P_X)\right]\underline{y}$$

Example 2. Let $\underline{y} = (y_1, \ldots, y_n)' \sim N(\underline{\mu}, I)$. Consider the mean $\overline{y} = (1/n)\sum_{i=1}^n y_i$ and the sum of squares $\sum_{i=1}^n (y_i - \overline{y})^2$.

(a) Show that $\overline{y}$ and $\sum_{i=1}^n (y_i - \overline{y})^2$ are independent.

Let $\underline{1} = (1, \ldots, 1)'$ be an $n \times 1$ vector of 1's. Then, $\overline{y} = (1/n)\underline{1}'\underline{y}$ and

$$\sum_{i=1}^n (y_i - \overline{y})^2 = (\underline{y} - \overline{y}\underline{1})'(\underline{y} - \overline{y}\underline{1}) = \underline{y}'\left(I - \underline{1}\,\underline{1}'/n\right)\underline{y},$$

and $I - \underline{1}\,\underline{1}'/n$ is a symmetric and idempotent matrix of rank $n-1$.

(b) Find the distribution of $\sum_{i=1}^n (y_i - \overline{y})^2$.

Let $\underline{\mu} = (\mu_1, \ldots, \mu_n)'$ and $\overline{\mu} = (1/n)\sum_{i=1}^n \mu_i$. By Theorem 3.16,

$$\sum_{i=1}^n (y_i - \overline{y})^2 \sim \chi^2\left[n-1, \frac{1}{2}\underline{\mu}'\left(I - \underline{1}\,\underline{1}'/n\right)\underline{\mu}\right], \text{ that is, } \chi^2\left[n-1, \frac{1}{2}\sum_{i=1}^n (\mu_i - \overline{\mu})^2\right].$$

Remark. Note that $\sum_{i=1}^n (y_i - \overline{y})^2 = \underline{w}'\underline{w}$, where

$$\underline{w} \equiv \left(I - \underline{1}\,\underline{1}'/n\right)\underline{y} \sim N\left[\left(I - \underline{1}\,\underline{1}'/n\right)\underline{\mu}, I - \underline{1}\,\underline{1}'/n\right].$$

By the corollary to Theorem 3.7, $\overline{y}$ and $\underline{w}$ are independent, implying $\overline{y}$ and $\sum_{i=1}^n (y_i - \overline{y})^2 = \underline{w}'\underline{w}$ are independent, as another way to show the independence of $\overline{y}$ and $\sum_{i=1}^n (y_i - \overline{y})^2$.

Idempotent matrices are of interest in our development because of Theorem 3.16. We now give three matrix lemmas (Lemmas 3.5 to 3.7) and then a useful theorem concerning idempotent matrices. The following lemma is used in proving Lemma 3.5.

Lemma 3.4. *(i) For two matrices A and B, if $\mathcal{R}(A)$ is a subspace of $\mathcal{R}(B)$ and* $\text{rank}(A) = \text{rank}(B)$*, then $\mathcal{R}(A) = \mathcal{R}(B)$. (ii) For two matrices A and B, if $\mathcal{C}(A)$ is a subspace of $\mathcal{C}(B)$ and* $\text{rank}(A) = \text{rank}(B)$*, then $\mathcal{C}(A) = \mathcal{C}(B)$.*

$\mathcal{C}(A) \subset \mathcal{C}(B)$

Lemma 3.5. *Let A be an $n \times n$ matrix. Then, A is idempotent if and only if* $\text{rank}(A) + \text{rank}(I - A) = n$.

$\Rightarrow$ if A is idempotent, then so is $I-A$

so $\text{rank}(A) + \text{rank}(I-A) = \text{tr}(A) + \text{tr}(I-A)$ ($\uparrow$ idempot)
$= \text{tr}(I) = n$

$\Leftarrow$ pick $\underline{x} \in \mathcal{C}(I-P_A)$, i.e. $\underline{x} = (I-P_A)\underline{q}$ for some $\underline{q}$

Then $A'\underline{x} = A'(I-P_A)\underline{q} = \underline{0}$ since $P_A A = A$ or $A' = A'P_A$

$\Rightarrow \underline{x} = (I-A')\underline{x}$ since $A'\underline{x} = \underline{0}$

$\Rightarrow \underline{x} \in \mathcal{C}(I-A')$

So $\mathcal{C}(I-P_A) \subset \mathcal{C}(I-A')$

& $\text{rank}(I-A') = \text{rank}(I-A)$ ($\uparrow$ transpose) $\overset{\text{assumption}}{=} n - \text{rank}(A)$
$= n - \text{rank}(P_A)$
$= n - \text{tr}(P_A)$
$= \text{tr}(I-P_A)$
$= \text{rank}(I-P_A)$ since P_A & $I-P_A$ are idempotent

By Lemma 3.4, $\mathcal{C}(I-P_A) = \mathcal{C}(I-A')$

$\Rightarrow (I-A') = (I-P_A)B$ for some B

$\Rightarrow (I-A)A = (I-A')'A = [(I-P_A)B]'A$
$= B'(I-P_A)A = 0$ since $P_A A = 0$

$\Rightarrow A$ is idempotent